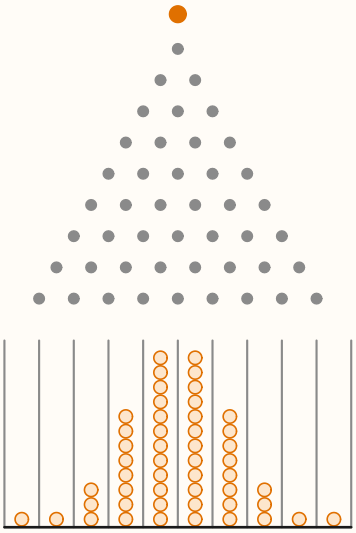
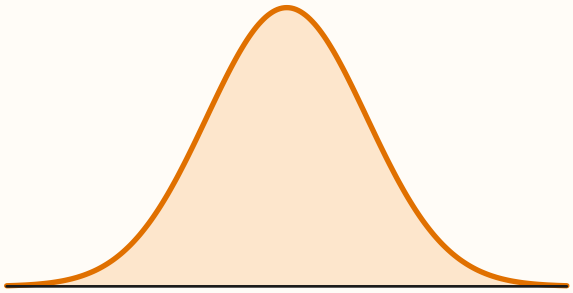
Chapter 26

The Normal distribution

why so many things pile up into a bell





the shape the balls are aiming for

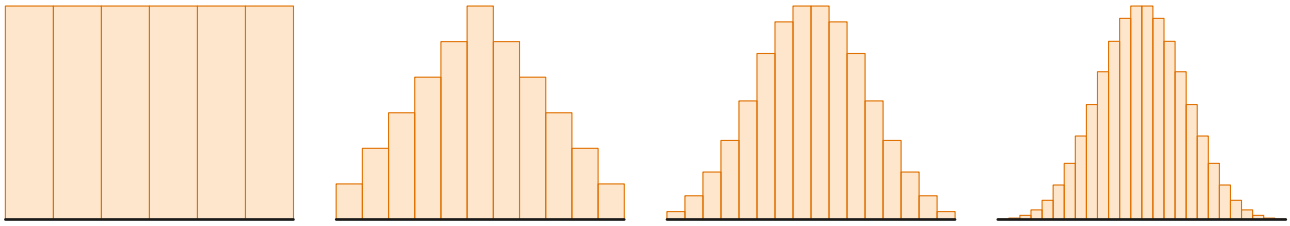
Drop balls into a board of pegs. At each peg a ball bounces left or right at random, yet the pile at the bottom always makes the same shape: a **bell**, high in the middle and fading at the edges. Heights, test scores and measurement errors make the same bell.

This chapter shows **why** the bell appears, how to read it (it's all about **area**), and how the **standard deviation** of Chapter 24 becomes a ruler for answering "what fraction of people...?" questions. You will never need to integrate the scary formula by hand.

Goes with **Lecture 9** (uniform vs Normal, sums of distributions, the bell curve equation, IQ, skewness and kurtosis, probability density, the seat-width problem).

/\ From flat to bell: add things together

Roll one die many times and every face turns up about equally: a **flat** histogram. Lecture 9 calls this **uniformly random** (coins, dice, roulette and lottery balls are all flat). A **distribution** is just the pattern of how often each value turns up.

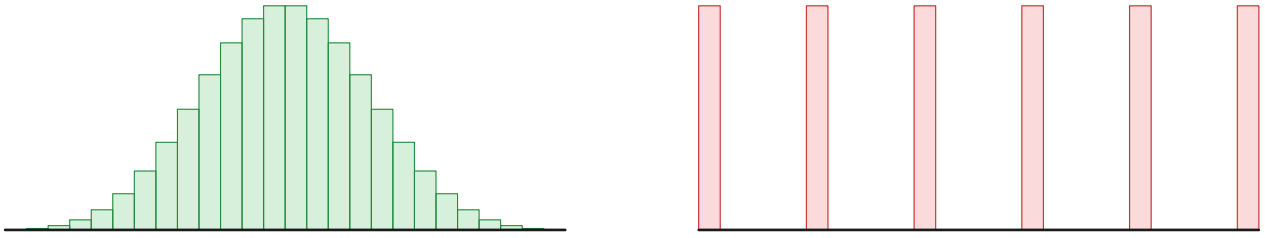


total of 1 die **total of 2 dice** **total of 3 dice** **total of 5 dice**

Now add two dice. A total of 7 can happen 6 ways but 2 only one way (the diagonals of Chapter 25's grid), so the histogram becomes a **triangle**. Three dice: a hump. Five: already a bell. Lecture 9 does the same with coins: one coin is flat; the number of heads in 100 coins is a beautiful bell (mean 50, standard deviation 5).

That's the secret: whenever a quantity is the **sum (or average) of many small random bits**, it comes out bell-shaped, whatever the bits looked like. Lecture 9: *"Any sum (or average) of multiple identically distributed distributions"* is Normal. Your height is the sum of thousands of small genetic and dietary nudges; that's why it's (roughly) Normal.

But not just any sum. The bits must be **separate** (independent, or only weakly linked). Both pictures below add up five dice numbers, but only one makes a bell:



5 separate rolls, added

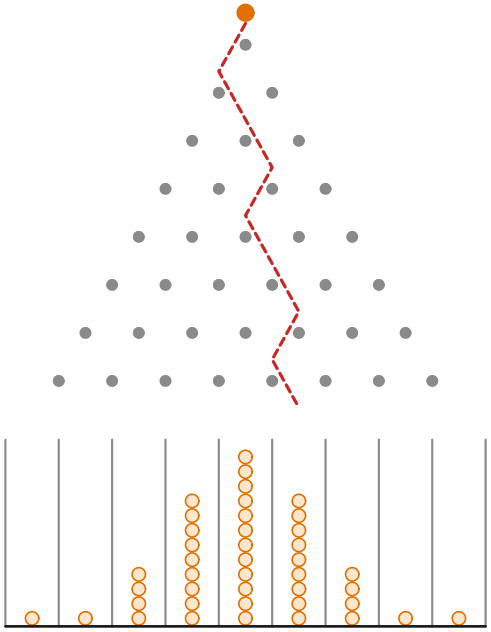
bell ✓

1 roll, copied 5 times, added

still flat: only 5, 10, ..., 30

Copying one roll adds no new randomness, so the shape is just the single die, stretched. The honest rule: a sum of **many** contributions is close to Normal when they're **independent or only weakly linked**, each has a **finite spread**, and **no single one dominates** the total. (If one huge nudge outweighs all the rest, the total just looks like that one nudge.)

The Galton board: coin flips you can watch



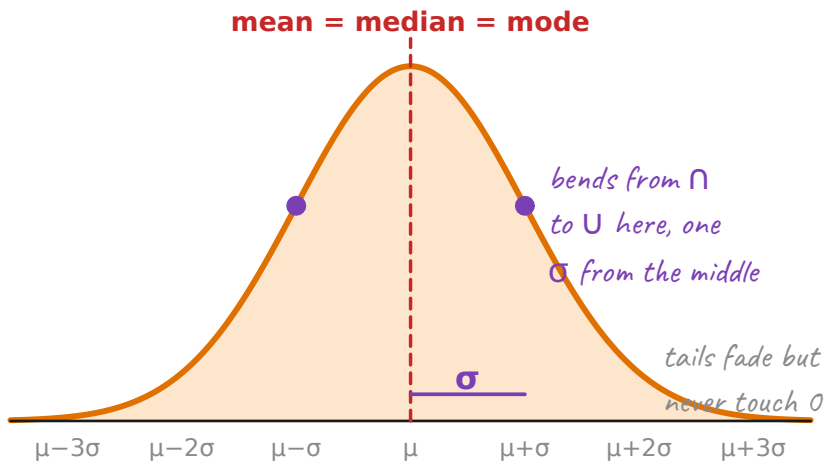
Each peg is a coin flip: left or right.
8 rows = 8 flips. The bin a ball lands in
= the number of rights.

The red path went L R R L R R L R:
5 rights, so bin 5.

Ending in the middle needs a mix of lefts
and rights, and there are many such paths.
The edges need all-left or all-right:
only one path each.

With 8 rows there are $2^8 = 256$ possible paths: 70 of them end in the middle bin but only 1 reaches each edge. Count paths, and the bell appears. (The counts 1, 8, 28, 56, 70, 56, 28, 8, 1 are the same numbers as expanding $(a+b)^8$.)

∧ Anatomy of the bell



Symmetric: the left half mirrors the right.

One peak in the middle.

Two numbers pin it down completely: the mean μ (where) and the standard deviation σ (how wide).

Lecture 9's equation of the curve looks terrifying, but read it piece by piece:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

(x − μ)/σ: how many σ's from the middle (the z-score, two pages on)

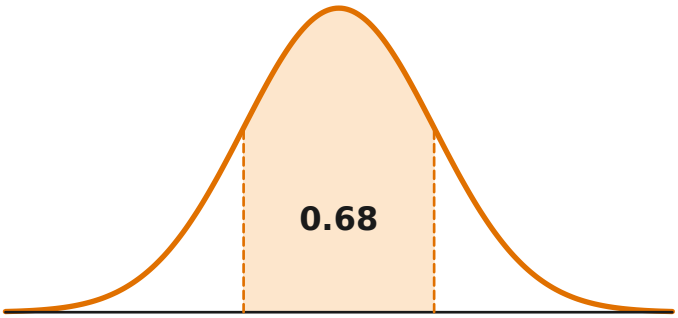
e to minus a square: 1 in the middle, shrinking fast either side (Chapter 21)

the front fraction just makes the total area exactly 1

With $\mu = 0$ and $\sigma = 1$ it's the **standard Normal**: $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. Every Normal curve is this one, slid along and stretched.

Probability is area

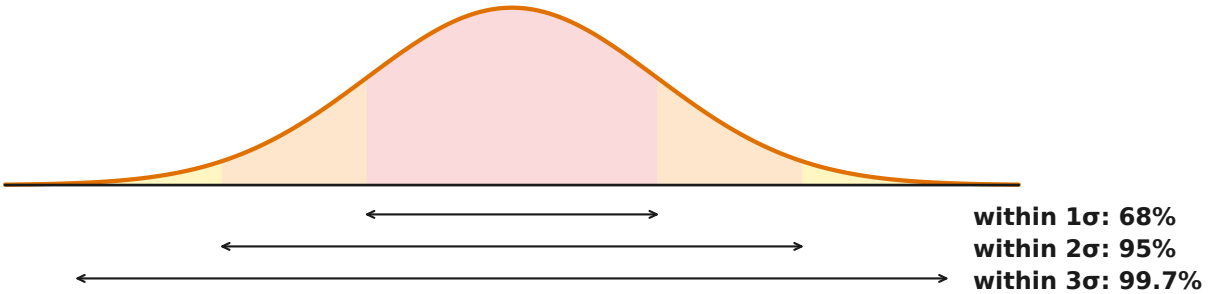
Imagine a histogram of 100,000 people's heights with ever thinner bars: the tops of the bars melt into the smooth curve. The height of the curve is called **probability density**. On its own a single height means little; **areas** are what count:



the whole area under the curve = 1 (everybody, 100%)

the shaded slice = the fraction of people between those values = the probability a random person lands there

The 68-95-99.7 rule. For **every** Normal distribution:



So about 2 in 3 people are within one standard deviation of the mean, 19 in 20 within two, and almost everyone within three.

Λ z-scores: a ruler marked in sigmas

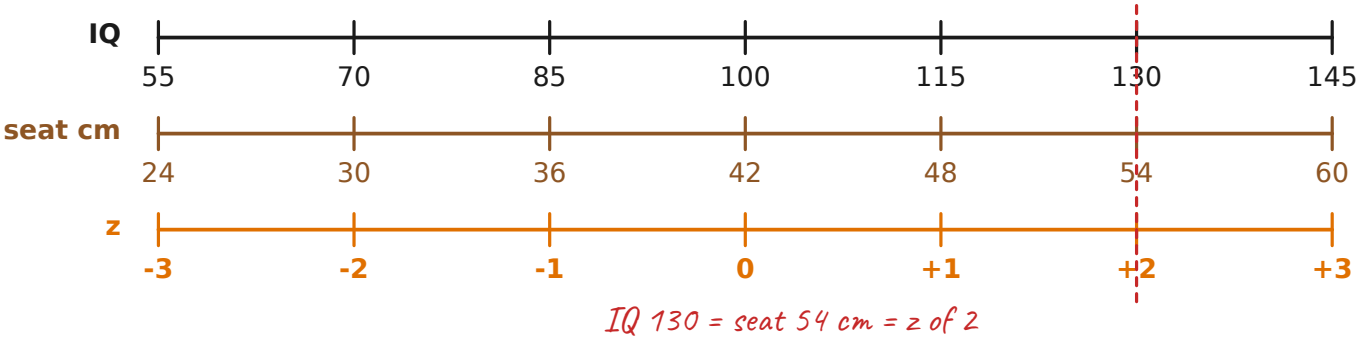
RULE: z-scores

$z = \dfrac{x - \mu}{\sigma}$: how many standard deviations x is above (+) or below (−) the mean.

Turned round: $x = \mu + z\sigma$.

Whatever μ and σ are, the z-scores follow the **standard Normal** (mean 0, standard deviation 1), so one table serves every Normal distribution.

IQ scores are Normal with mean 100 and standard deviation 15 (Lecture 9). Lay a **second ruler** under the IQ ruler, marked in steps of one σ. The reading on that ruler is the **z-score**: **how many standard deviations above (+) or below (−) the mean**.



$$z = \frac{x - \mu}{\sigma}$$ ← distance from the mean ÷ the size of one step

$$\text{IQ } 75: \quad z = \frac{75 - 100}{15} = \frac{-25}{15} \approx -1.67$$ ← 1.67 σ's below

Once everything is in z-units, **one** table (or calculator button) works for every Normal distribution in the world. That table gives $\Phi(z)$ ("phi of z"): the area to the **left** of z, i.e. the fraction of people **below** that point.

z = 0	z = 0.5	z = 1	z = 1.28	z = 1.33	z = 1.645	z = 1.96	z = 2	z = 2.326	z = 3
0.5000	0.6915	0.8413	0.8997	0.9082	0.9500	0.9750	0.9772	0.9900	0.9987

Areas from the table: the four rules

RULE: areas under a Normal curve

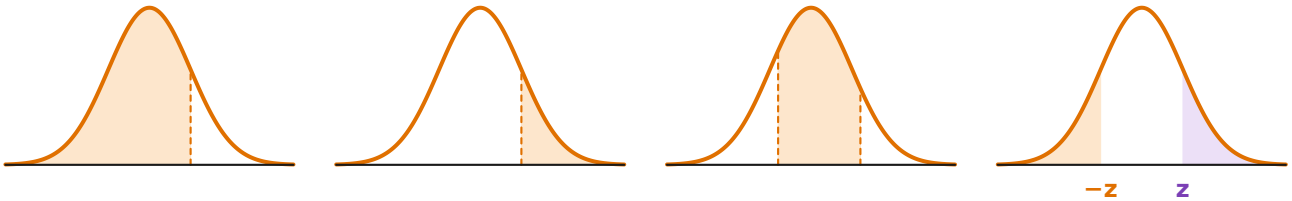
$\Phi(z)$ = the area to the **left** of z. So $p(X < x) = \Phi(z)$.

Above a value: $p(X > x) = 1 - \Phi(z)$, because the whole area is 1.

Between two values: $p(a < X < b) = \Phi(z_b) - \Phi(z_a)$.

Negative z: printed tables only give positive z. By symmetry $\Phi(-z) = 1 - \Phi(z)$. Also $\Phi(0) = 0.5$.

For a smooth curve, $<$ and $\leq$ give the same area: the chance of hitting one exact value is 0.



$p(X < x) = \Phi(z)$ $\quad$ $p(X > x) = 1 - \Phi(z)$ $\quad$ $\Phi(z_b) - \Phi(z_a)$ $\quad$ $\Phi(-z) = 1 - \Phi(z)$

PROCEDURE: finding a Normal probability

1. **Standardise**: turn each x into $z = \frac{x-\mu}{\sigma}$.
2. **Sketch** the bell, mark the mean and your value(s), and shade the area the question asks for.
3. **Look up** Φ for each z (for a negative z use $1 - \Phi(|z|)$).
4. **Finish**: left area, done; right area, $1 - \Phi$; between, larger Φ minus smaller Φ. Check against the sketch: should the answer be more or less than 0.5?

Example: IQ is Normal with $\mu = 100$, $\sigma = 15$. What fraction score **above** 120?

$$z = \frac{120 - 100}{15} = 1.33$$ ← standardise (2 d.p.)

$$p(X > 120) = 1 - \Phi(1.33) = 1 - 0.9082 = 0.0918$$

right of z: 1 − Φ

Example: what fraction score **below** 85?

$$z = \frac{85 - 100}{15} = -1$$ ← standardise

$$\Phi(-1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$$

negative z: symmetry

Example: what fraction score **between** 85 and 120?

$$\Phi(1.33) - \Phi(-1) = 0.9082 - 0.1587 = 0.7495$$

larger minus smaller: about 75%

✗ Common mistakes

✗ $p(X > x) = \Phi(z)$ ✓ $p(X > x) = 1 - \Phi(z)$

Φ always gives the area to the LEFT

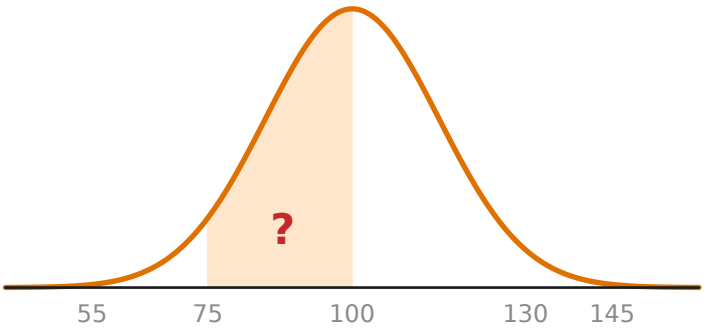
✗ $z = \frac{x - \mu}{\sigma^2}$ ✓ $z = \frac{x - \mu}{\sigma}$

divide by the standard deviation, not the variance

✗ $\Phi(-1) = -0.8413$ ✓ $\Phi(-1) = 1 - 0.8413 = 0.1587$

an area is never negative: use the symmetry rule

Lecture 9's IQ range: what fraction scores 75 to 100?



Area between 75 and 100
= (area left of 100)
− (area left of 75)

$$z_{100} = 0, \quad z_{75} = -1.667$$ ← *convert both ends to z*

$$\Phi(-1.667) = 1 - \Phi(1.667) = 1 - 0.9522 = 0.0478$$

negative z: use the symmetry rule

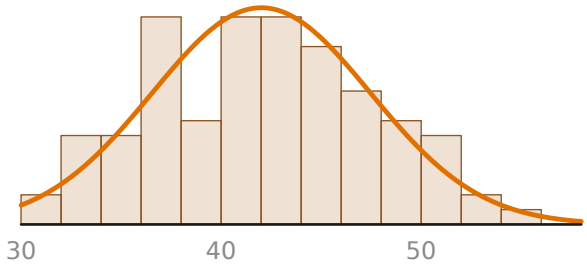
$$\Phi(0) - \Phi(-1.667) = 0.5000 - 0.0478$$ ← *half the area is left of the middle*

$$= 0.4522 \approx 45\%$$ ← *exactly the lecture's figure*

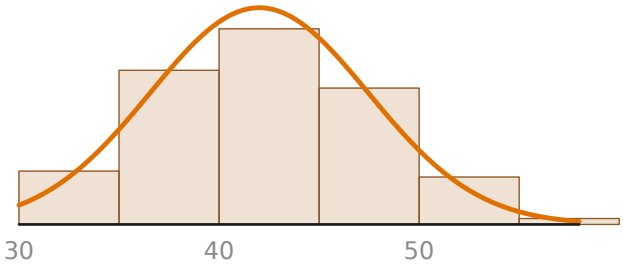
(Lecture 9 uses this to show how a far-right group targeted its propaganda at people with IQ 75-100. The maths is neutral; what people do with it isn't.)

The aeroplane seat problem

Lecture 9: design seats wide enough for **90% of people**; the widest 10% buy two tickets. First, measure a sample. The slides list 100 seat widths (cm); here they are as histograms, binned two ways:



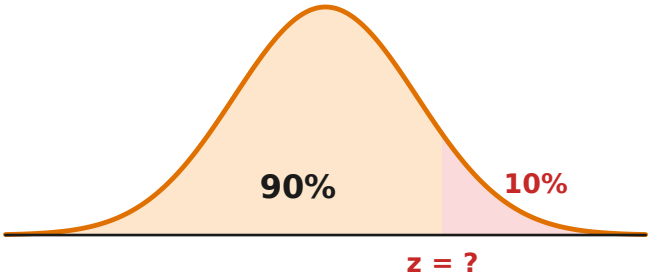
2 cm bins



5 cm bins

Same data, different binning, different look (the lecture's point). Either way, the orange Normal curve fits well. The data give mean **42.0 cm** and standard deviation **5.46 cm**; the slides round this to **6 cm**, so we'll use 6.

Now work backwards. Before, we had x and wanted the area. Now we have the area (90% to the left) and want x:



Find the z with Φ(z) = 0.90.
The table (previous page):
Φ(1.28) ≈ 0.90, so z ≈ 1.28.
Two tickets beyond here.

$$z = \frac{x - \mu}{\sigma} \Rightarrow x = \mu + z\sigma$$ ← *transpose (Chapter 14)*

$$x = 42 + 1.28 \times 6 = 49.7 \text{ cm}$$

Seats 49.7 cm wide fit 90% of passengers. (With the unrounded 5.46 cm you'd get 49.0 cm; rounding σ costs airlines 7 mm per seat.)

Working backwards: cut-offs, μ and σ

PROCEDURE: inverse Normal problems

1. Turn the percentage into an area to the **left**: bottom 10% is 0.10; top 10% is 0.90; the middle 90% leaves 5% in each tail, so 0.05 and 0.95.
2. Find the z with that $\Phi(z)$: read the table backwards, or use a calculator's inverse Normal. A left-tail area below 0.5 gives a **negative** z.
3. Put z into $z = \frac{x-\mu}{\sigma}$ and solve for the missing quantity: $x = \mu + z\sigma$, $\mu = x - z\sigma$, or $\sigma = \dfrac{x-\mu}{z}$.
4. If μ and σ are **both** unknown you need two facts: write $x = \mu + z\sigma$ for each and solve them as simultaneous equations (Chapter 15).

area to the left, Φ(z)	z	use it for
0.90	1.2816	top 10% (or bottom 90%)
0.95	1.6449	top 5%; middle 90% is ±1.645
0.975	1.96	top 2.5%; middle 95% is ±1.96
0.99	2.3263	top 1%; middle 98% is ±2.326
0.10	−1.2816	bottom 10%: the mirror image of 0.90

Example: IQ ($\mu = 100$, $\sigma = 15$): what score cuts off the **bottom** 10%?

$$\Phi(z) = 0.10 \;\Rightarrow\; z = -1.2816$$ ← a bottom tail: negative z

$$x = 100 + (-1.2816) \times 15 = 80.8$$ ← $x = \mu + z\sigma$

Example: between which two IQ scores do the **middle** 90% lie?

$$z = \pm 1.645, \qquad 100 \pm 1.645 \times 15 = 100 \pm 24.7$$

5% in each tail

So from **75.3** to **124.7**.

Example: a machine fills bags with $\sigma = 5$ g. Only 2.5% of bags may weigh under 500 g. What mean μ should it be set to?

$$\Phi(z) = 0.025 \;\Rightarrow\; z = -1.96$$

$$500 = \mu - 1.96 \times 5 \;\Rightarrow\; \mu = 509.8 \text{ g}$$ ← $\mu = x - z\sigma$

Example: exam marks have mean 60 and 10% of students score over 80. Find σ.

$$\Phi(z) = 0.90 \;\Rightarrow\; z = 1.2816$$ ← top 10% means 90% to the left

$$\sigma = \frac{80 - 60}{1.2816} = 15.6$$ ← $\sigma = (x - \mu)/z$

✗ Common mistakes

✗ $\text{top } 10\% : \; z = -1.28$

the top tail is above the mean

✓ $\text{top } 10\% : \; z = +1.28$

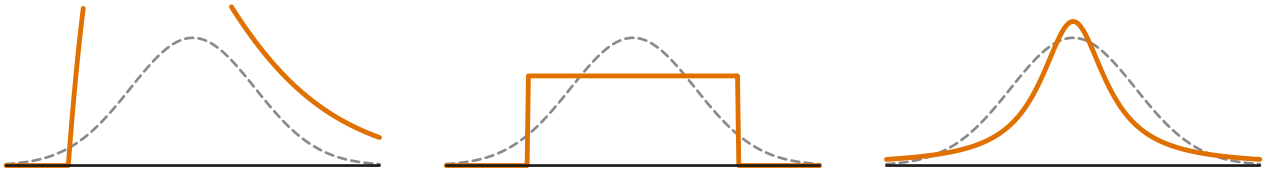
✗ $\text{middle } 90\% : \; \Phi(z) = 0.90$

the 10% left over is split between two tails

✓ $\text{middle } 90\% : \; \Phi(z) = 0.95$

Is it Normal? And stay critical

Probably Normal (Lecture 9): averaged qualities like IQ, test scores, rainfall, molecular speeds, football crowds, women's heights. **Definitely not:** anything **lopsided** or with a **hard limit on one side**, like wages (skewed, Chapter 24). Two numbers measure the damage:



skewness ≠ 0
lopsided

kurtosis < 0
too flat (e.g. uniform)

kurtosis > 0
too peaky, heavy tails

A Normal distribution has skewness 0 and kurtosis 0. Lecture 9's table: one uniform variable has kurtosis −1.18 (flat), the sum of four has −0.32: **adding makes it more Normal**, the lesson of page 2.

Be critical. Lecture 9 tells the story of *The Bell Curve* (1994) and a world IQ map built from biased samples: one country's "national IQ" came from a group of children in a Spanish care home; inconvenient high-scoring samples were dropped. Chapter 23's detective questions (who was sampled? what's missing?) matter more than the maths. The lecture's last word: **scientists can be lazy, prejudiced and wrong but still get published, so develop your own analytical skills**.

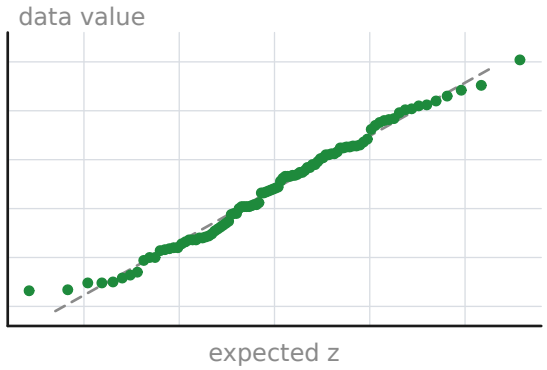
Checking for Normality: the probability plot

RULE: normal probability plot

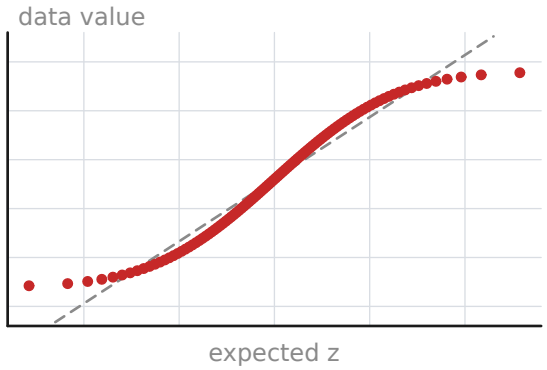
Sort the data. Plot each value against the z-score it **would** have if the data were exactly Normal (the kth of n values goes with the z that has $\Phi(z) = \frac{k-0.5}{n}$).

Normal data lie close to a **straight line**. Ends that bend away from the line mean skew, or tails that are too light or too heavy.

Software (Minitab, in Lecture 9) draws it for you: your job is to judge whether the points are straight. Lecture 9: the difference from a uniform data set is much clearer here than in a histogram.



the 100 seat widths: straight, so Normal



uniform data: bent ends, not Normal

Is human height Normal? Lecture 9 lists **women's** heights as roughly Normal, but **people's** heights (men and women together) as not: the mix is two overlapping bells with different means, which is wider and flatter on top than one Normal. Even one group's heights are only **approximately** Normal: the curve's tails go on for ever, but nobody is 50 cm or 3 m tall. A model can be very useful without being exactly true.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

z-score	$z = \dfrac{x - \mu}{\sigma}$ how many standard deviations from the mean
Back from z	$x = \mu + z\sigma, \qquad \mu = x - z\sigma, \qquad \sigma = \dfrac{x - \mu}{z}$
Below a value	$p(X < x) = \Phi(z)$ Φ is the area to the left
Above a value	$p(X > x) = 1 - \Phi(z)$
Between two values	$p(a < X < b) = \Phi(z_b) - \Phi(z_a)$
Negative z	$\Phi(-z) = 1 - \Phi(z), \qquad \Phi(0) = 0.5$ tables only give positive z
68-95-99.7 rule	within 1σ: 68%; within 2σ: 95%; within 3σ: 99.7%
Normal probability	standardise, sketch and shade, look up, then complement or subtract
Inverse Normal	percentage → area to the left → z (table backwards) → x, μ or σ
Useful z values	Φ = 0.90: z = 1.2816; 0.95: 1.645; 0.975: 1.96; 0.99: 2.326. Middle 90%: ±1.645; middle 95%: ±1.96
Normal curve	$f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ total area 1; symmetric about μ; mean = median = mode
When is it Normal?	sums or averages of many small independent effects; check with a probability plot (straight line)

Λ Your turn

Use the Φ table on the z-score page. Answers in grey.

z-scores

1. IQ 115: z = ?
2. IQ 70: z = ?
3. A seat width of 36 cm (μ = 42, σ = 6): z = ?
4. What IQ has z = 1.645?

1. 1 2. −2 3. −1 4. 100 + 1.645 × 15 ≈ 124.7

Areas

5. Fraction of people with IQ below 130
6. Fraction with IQ above 115
7. Fraction between IQ 85 and 115
8. Fraction with IQ between 100 and 124.7

5. Φ(2) = 0.977 6. 1 − Φ(1) = 0.159 7. 0.683 (the 68% rule) 8. 0.95 − 0.5 = 0.45

Backwards

9. An exam is Normal, μ = 60, σ = 10. The top 10% get a distinction. What mark is needed?
10. Which is more Normal: one die, or the average of 20 dice? Why?

9. 60 + 1.28 × 10 ≈ 72.8 10. the average of 20: sums of many independent bits become bell-shaped

Find the unknown

11. IQ (μ = 100, σ = 15): which score cuts off the bottom 10%?
12. Between which two IQ scores do the middle 95% lie?
13. Bags have σ = 4 g, and 5% weigh under 250 g. Find μ.
14. Exam marks: μ = 55, and 10% score above 75. Find σ.
15. A value of 70 has z = 2 and a value of 40 has z = −1. Find μ and σ.
16. Heights: μ = 170 cm and p(X > 185) = 0.0668. Find σ.

11. Φ(z) = 0.10, z = −1.2816; 100 − 1.2816 × 15 ≈ 80.8 12. z = ±1.96; 100 ± 29.4: 70.6 to 129.4 13. z = −1.645; μ = 250 + 1.645 × 4 ≈ 256.6 g 14. z = 1.2816; σ = (75 − 55)/1.2816 ≈ 15.6 15. 70 = μ + 2σ and 40 = μ − σ, so σ = 10, μ = 50 16. Φ(z) = 0.9332, z = 1.5; σ = 15/1.5 = 10